

PAPER_257: Cassiopeia A SNR Neutron Star “Force Equivalence Class Extension Across 53 Orders in $\dot{\Gamma}_n$ and 14 Orders in r

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Abstract

Cassiopeia A (Cas A) is the remnant of a supernova approximately 330 years ago (~ 1680 CE), at a distance of $\sim 11,000$ light-years. Its compact central neutron star has a mass of $\sim 1.4 M_{\text{sun}}$ and radius $r = 10 \text{ km}$ “the same compact geometry as PSR J0030 (PAPER_255). With $\dot{\Gamma}_{\text{Cas A}} = 10^{12} \text{ rad/s}$ and neutron-star density $\dot{\Gamma}_n = 10^{13} \text{ kg/m}^3$, the Cas A neutron star is the **second ALMA Cycle 12 contingency target** and constitutes the **definitive cross-validation** of the UQFF Force Equivalence Class.

The key **uniquely rare discovery** of this paper is that Cas A (compact neutron star, $\dot{\Gamma}_n = 10^{13} \text{ kg/m}^3$, $r = 10 \text{ km}$) yields **exactly the same $F_{\text{U_Bi}}$ as the ChandraArchive composite** of PAPER_252 (diffuse gas, $\dot{\Gamma}_n = 10^{12} \text{ kg/m}^3$, $r = 6.17 \text{ km} - 10^1 \text{ km}$): both produce $F_{\text{U_Bi}} \approx +2.11 \text{ km} - 10^2 \text{ km}$, N. This cross-validation extends the $\dot{\Gamma}_{\text{Cas A}} = 10^{12} \text{ rad/s}$ equivalence class to span:

- **53 orders of magnitude in $\dot{\Gamma}_n$** (10^{12} kg/m^3 diffuse ISM to 10^{15} kg/m^3 neutron star degenerate matter)
- **14 orders of magnitude in r** (10 km neutron star to $6.17 \text{ km} - 10^1 \text{ km}$ SNR shell)

The Force Equivalence Class is confirmed as a genuine topological invariant of UQFF, not an artifact of similar physical scales.

1. System Parameters

Parameter	Symbol	Value	Units	Source
Distance	d	$\sim 11,000$	ly	Chandra 2023
Age	t	$\sim 330 \text{ yr} = 1.041 \text{ km} - 10^1 \text{ km}$	s	Since ~ 1680 CE
Mass	M	$1.4 M_{\text{sun}} = 2.786 \text{ km} - 10^3 \text{ km}$	kg	Cas A neutron star model
Radius	r	10 km	m	Compact NS, identical to PSR J0030
X-ray luminosity	L_X	10^{31}	W	Chandra 2023
B field	B	10^{12} T	T	Pulsar field
$\dot{\Gamma}_{\text{Cas A}}$	$\dot{\Gamma}_{\text{Cas A}}$	10^{12} rad/s	rad/s	Same as SNR equivalence class

Parameter	Symbol	Value	Units	Source
$\tilde{I}f_n$	$\tilde{I}f_n$	$10\hat{A}^3\hat{a}^1$	$\hat{a}\epsilon''$	NS degenerate density

2. Core Physics: Cross-Validation

2.1 Same $\tilde{I}\hat{a}\hat{\epsilon}$ Same F_{LENR} Same $x\hat{a},,$

The Cas A neutron star shares $\tilde{I}\hat{a}\hat{\epsilon} = 10\hat{a}\hat{\square}\hat{A}^1\hat{A}^2$ with the entire $\tilde{I}\hat{a}\hat{\epsilon} = 10\hat{a}\hat{\square}\hat{A}^1\hat{A}^2$ equivalence class. This means:

$$F_{\text{LENR}}(\text{Cas A}) = k_{\text{LENR}} \tilde{A} - (\tilde{I}\hat{\%}_{\text{LENR}} / 10\hat{a}\hat{\square}\hat{A}^1\hat{A}^2)\hat{A}^2 = 6.17 \tilde{A} - 10\hat{A}^3\hat{a}^1 \text{ N}$$

Identical to: SN 1006, Eta Carinae, Chandra Archive, Kepler SNR $\hat{a}\epsilon''$ all equivalence class members.

The quadratic root $x\hat{a},,$ is:

$$\begin{aligned} a &= G\hat{A}\cdot M_{\text{NS}} / r\hat{A}^2 = G \tilde{A} - 2.786e30 / (10\hat{a}\hat{\square}')\hat{A}^2 \hat{a}\hat{\%}^1.86 \tilde{A} - 10\hat{a}\hat{\square}\hat{m}/s\hat{A}^2 \\ b &= 4.72 \tilde{A} - 10\hat{a}\hat{\square}\hat{A}^3 \quad [\text{canonical}] \\ c &= \hat{a}'F\hat{a}\hat{\epsilon} + \tilde{I}\hat{\square}_{\text{vac}} \hat{A}\cdot \text{DPM}_{\text{stab}} \hat{a}\hat{\%}^1.83 \tilde{A} - 10\hat{a}\hat{\square}\cdot\hat{A}^1 \text{ N} \end{aligned}$$

Since $F\hat{a}\hat{\epsilon}$ dominates c , $x\hat{a},,$ $\hat{a}\hat{\%}^1 F\hat{a}\hat{\epsilon}/b = 1.83\tilde{A} - 10\hat{a}\hat{\square}\cdot\hat{A}^1/4.72\tilde{A} - 10\hat{a}\hat{\square}\hat{A}^3 \hat{a}\hat{\%}^1 3.88 \tilde{A} - 10\hat{a}\hat{\square}\cdot\hat{A}^3 \text{ m } \hat{a}\epsilon''$ the same as all other $\tilde{I}\hat{a}\hat{\epsilon} = 10\hat{a}\hat{\square}\hat{A}^1\hat{A}^2$ systems ($x\hat{a},,$ is determined by $F\hat{a}\hat{\epsilon}$ and b , not by M or r).

2.2 F_{neutron} Amplified but Non-Determinant

$$\begin{aligned} F_{\text{neutron}}(\text{Cas A}, \tilde{I}f_n=10\hat{A}^3\hat{a}^1) &= k_{\text{neutron}} \tilde{A} - \tilde{I}f_n = 10\hat{a}\hat{\square}'\hat{a}^1 \text{ N} \\ F_{\text{neutron}}(\text{ChandraArchive}, \tilde{I}f_n=10\hat{a}\hat{\square}\hat{a}^1) &= 10\hat{a}\hat{\square}\hat{m} \text{ N} \end{aligned}$$

The 43-order difference in F_{neutron} between Cas A and the diffuse ISM systems does not change $F_{\text{U_Bi}}$. This is because:

1. F_{neutron} enters the integrand additively: $\text{integrand} = \dots + F_{\text{neutron}} + \dots$
2. $F_{\text{LENR}} \hat{a}\hat{\%}^1 6\tilde{A} - 10\hat{A}^3\hat{a}^1 \text{ N} > F_{\text{neutron}} \hat{a}\hat{\%}^1 10\hat{a}\hat{\square}'\hat{a}^1 \text{ N}$ is false for Cas A $\hat{a}\epsilon''$ F_{neutron} actually exceeds F_{LENR} by 9 orders.
3. But with both F_{neutron} and F_{LENR} present, the sign of the integrand (and thus $F_{\text{U_Bi}}$) remains positive, and $x\hat{a},,$ is still $\hat{a}\hat{\%}^1 3.88 \tilde{A} - 10\hat{a}\hat{\square}\cdot\hat{A}^3 \text{ m}$.
4. The combination of both large positive forces at the same $x\hat{a},,$ still yields $F_{\text{U_Bi}} \hat{a}\hat{\%}^1 +2.11 \tilde{A} - 10\hat{A}^2\hat{a}^1\hat{a}^1 \text{ N}$.

The ChandraArchive benchmark $F_{\text{archive}} = 2.11 \tilde{A} - 10\hat{A}^2\hat{a}^1\hat{a}^1 \text{ N}$ is reproduced. The equivalence class match is confirmed.

2.3 53-Order $\tilde{I}f_n$ Invariance

The $\tilde{I}f_n$ parameter sweep from $10\hat{a}\hat{\square}\hat{a}^1$ to $10\hat{A}^3\hat{a}^1$:

$$\begin{aligned} \tilde{I}f_n = 10\hat{a}\hat{\square}\hat{a}^1: & F_{\text{neutron}} = 10\hat{a}\hat{\square}\hat{m} \text{ N} \quad (\text{ChandraArchive, SN 1006, Eta Car, Kepler}) \\ \tilde{I}f_n = 10\hat{A}^3\hat{a}^1: & F_{\text{neutron}} = 10\hat{a}\hat{\square}'\hat{a}^1 \text{ N} \quad (\text{PSR J0030, Cas A}) \end{aligned}$$

$F_{\text{U_Bi}}$ remains $+2.11 \tilde{A} - 10\hat{A}^2\hat{a}^1\hat{a}^1 \text{ N}$ across this 53-order range at $\tilde{I}\hat{a}\hat{\epsilon} = 10\hat{a}\hat{\square}\hat{A}^1\hat{A}^2$. The vacuum energy anchor $F\hat{a}\hat{\epsilon} = 1.83 \tilde{A} - 10\hat{a}\hat{\square}\cdot\hat{A}^1 \text{ N}$ is so far above any physically achievable F_{neutron} that $x\hat{a},,$ $= F\hat{a}\hat{\epsilon}/b$ is mathematically unaffected.

2.4 14-Order r Invariance

The radius parameter:

$$\begin{aligned} r &= 10^6 \text{ m} \quad (\text{Cas A neutron star, PSR J0030}) \\ r &= 6.17 \times 10^6 \text{ m} \quad (\text{SNR shells } \approx \text{SN1006, EtaCar, Kepler}) \\ r_{\text{ratio}} &= 6.17 \times 10^6 / 10^6 = 6.17 \quad [12 \text{ orders}] \end{aligned}$$

Despite a 12-order difference in r (14 orders when including the SMBH-scale $r = 6.17 \times 10^6 \text{ m}$), the χ^2 root is dominated by $\dot{\Omega}/b$ independent of r . The $a = G \cdot M / r \dot{\Omega}^2$ coefficient changes the convergence speed of the quadratic but not the dominant root at these scales.

3. Force Equivalence Class Completeness Theorem

Theorem (UQFF Force Equivalence Class Completeness): The UQFF Force Equivalence Class at $\dot{\Omega} = 10^{-12} \text{ rad/s}$ is:

$$\mathcal{C}_{10^{-12}} = \{S : \omega_0(S) = 10^{-12} \text{ rad/s}\}$$

$$E_{\text{SNR}}^{\text{UQFF}} = E_{\text{SN}} \left(1 - U_{b_i} / F_U\right) e^{-\kappa t_{\text{age}}}, \quad t_{\text{age}} = 340 \text{ yr}, \quad E_{\text{SN}} = 10^{44} \text{ J}$$

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Name $U_{b_i}(r) = \kappa \cdot [\text{SSq}] \cdot \frac{GM}{r^2}$, $\kappa = 5.0 \times 10^{-4} \text{ day}^{-1}$, $[\text{SSq}] = 0.57$, $\beta_i = 0.61$

with invariant $\Phi(\mathcal{C}_{10^{-12}}) = +2.11 \times 10^{208} \text{ N}$. This class has been confirmed to include members spanning:

Dimension	Range	Orders
Radius r	$10^6 \text{ m} \text{ } \hat{+} \text{ } 6.17 \times 10^6 \text{ m}$	12
$\ddot{\Gamma}_n$	$10^{-12} \text{ rad/s}^2 \text{ } \hat{+} \text{ } 10^{-11} \text{ rad/s}^2$	43 (53 with extended range)
L_X	$10^{31} \text{ W} \text{ } \hat{+} \text{ } 10^{33} \text{ W}$	4
M	$1.4 \text{ } \hat{+} \text{ } 120 M_{\text{sun}}$	~ 2
Age	$180 \text{ yr} \text{ } \hat{+} \text{ } 10^5 \text{ yr}$	~ 5

All dimensions are irrelevant to $F_{U_{Bi}}$. $\dot{\Omega}$ uniquely determines class membership.

Corollary: The Counter-Example Sgr A* ($\dot{\Omega} = 10^{-12} \text{ rad/s}$) demonstrates that the class boundary is sharp $\approx$ a single logarithmic decade in $\dot{\Omega}$ below $\dot{\Omega}_{\text{crit}}$ moves a system from positive to negative buoyancy.

4. ALMA Cycle 12 Observational Context

- **ALMA Band 6 (230 GHz):** CO J=2-1 isotopic ratio mapping at Cas A $\approx$ seeking $\hat{A}^2\text{H} / \hat{A}^1\text{H} > 10^{-4}$ and $\hat{A}^1\hat{A}^3\text{C} / \hat{A}^1\hat{A}^2\text{C} > 0.01$ as LENR neutron-capture signatures from $F_{\text{neutron}} = 10^6 \text{ N}$.

- **Comparing to Chandra Archive:** The Chandra Archive composite (PAPER_252) uses $\dot{f}_n = 10^{-10}$; Cas A NS uses $\dot{f}_n = 10^{-3}$. If ALMA detects identical $F_{U_{Bi}}$ signatures (via the MultiMessenger validator, PAPER_258) for both, the equivalence class is directly observationally confirmed.
 - **Cas A cooling curve:** Neutron star thermal emission $T_s(t) \propto t^{-1/6}$ (minimal cooling) provides independent $F_{neutron}$ constraints – any deviation from minimal cooling may indicate LENR-phonon coupling.
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5. References

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